

CSCN8020 Assignment 3

Deep Q-Network Control of the Unitree G1 Left Elbow

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Project Summary

This project implements a student-written PyTorch Deep Q-Network to control the Unitree G1 humanoid robot's left elbow in MuJoCo. The agent observes elbow angle, velocity, goal angle, and signed error, then selects decrease, hold, or increase actions. Separate online and target networks, experience replay, epsilon-greedy exploration, Huber loss, and checkpoint loading are implemented directly. Two epsilon-decay schedules were compared under controlled conditions. Both trained agents achieved 20 out of 20 successful greedy evaluation episodes across four target angles. Configuration A was selected and compared fairly with the supplied rule-based policy. Reproducible code, metrics, plots, checkpoint, report, and execution instructions are included.

GitHub Repository

https://github.com/chence/CSCN8020_Assignment3

Cloneable URL

https://github.com/chence/CSCN8020_Assignment3.git